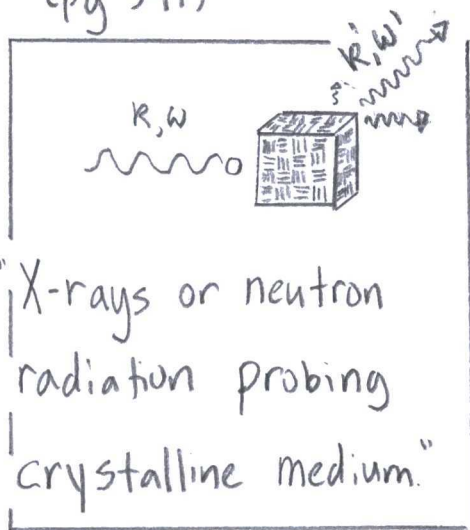


# Chapter 7: Response Functions:

Exercise #1:  
(pg 391)



(from exercise) "Response Function"

$$\chi(k, k'; \omega) \propto \delta_{k-k'-\epsilon}$$

(7.3) "Fourier transform for temporal convolution"

$$\chi_{ij}(r, \omega) = \int d^d r' \chi_{ij}(r, r'; \omega) F_j'(r', \omega)$$

$$\chi(k, \omega) = \left( \frac{d^d}{d^d r'} \right) \int d^d r' \chi_{ij}(k, k'; \omega) F_j'(k', \omega)$$

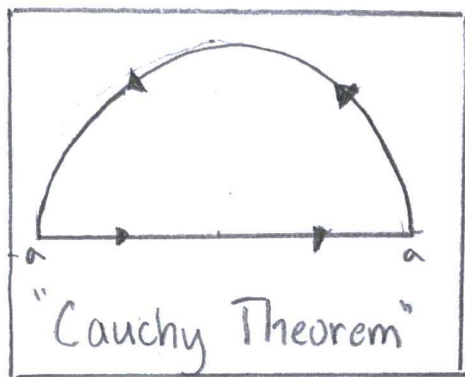
$$= \chi_{ij}(k, k'; \omega) F_j'(k', \omega)$$

$$\propto \delta_{k-k'-g} \cdot F_j'(k', \omega)$$

$$\propto F_j'(k', \omega)$$

Response  $\propto$  Forces in lattice

Exercise #2:  
(pg 399)



(7.23) "Master Function"

$$C(z) = \frac{1}{z} \sum_{\alpha, \beta} \frac{\chi_{1, \alpha \beta} \chi_{2, \beta \alpha}}{z + \Xi_{\alpha, \beta}} \left[ e^{-\beta \Xi_{\alpha}} - \zeta e^{-\beta \Xi_{\beta}} \right]$$

where  $\Xi$  = "Total Potential"  
 $= E - \mu \cdot N$

$Z$  : "General complex variable"

$X_{\alpha\beta}$  = "Expected value"

$$= \langle \Psi_{\alpha} | \hat{x} | \Psi_{\beta} \rangle$$

If  $Z = w + i\eta$ ,  $\eta$  = convergent factor

$$C(t) = \frac{1}{2\pi} \int dw e^{-i\omega t} \cdot C(w \pm i\eta)$$

$$= \frac{1}{2\pi} \int dw e^{-i\omega t} \cdot \left( \frac{1}{Z} \sum \frac{X_{1\alpha\beta} X_{2\beta\alpha}}{Z + \Xi_{\alpha\beta}} \left[ e^{-\beta \Xi_{\alpha}} - \xi e^{-\beta \Xi_{\beta}} \right] \right)$$

$$= \frac{X_{1\alpha\beta} X_{2\beta\alpha} [e^{-\beta \Xi_{\alpha}} - \xi e^{-\beta \Xi_{\beta}}]}{2\pi \cdot Z} \int dw \frac{e^{-i\omega t}}{w \pm i\eta + \Xi_{\alpha\beta}}$$

$$= \frac{X_{1\alpha\beta} X_{2\beta\alpha} [e^{-\beta \Xi_{\alpha}} - \xi e^{-\beta \Xi_{\beta}}]}{2\pi Z} \cdot e^{(\mp\eta + i\Xi)t} \int du \frac{e^{-u}}{u}$$

u-substitution:

$$u = i(\omega t \pm (i\eta + \Xi_{\alpha\beta})t)$$

$$\frac{du}{d\omega} = i t$$

Exponential Integral:

$$Ei(x) = - \int_{-x}^{\infty} \frac{e^{-t}}{t} dt = \int_{-\infty}^x \frac{e^t}{t} dt$$

$$= \frac{-\chi_{\alpha\beta} \chi_{\beta\alpha} \left[ e^{-\beta \Xi_{\alpha}} e^{-\beta \Xi_{\beta}} \right]}{2\pi \cdot Z} \cdot e^{(\pm\eta + i\Xi)t}$$

$$\times F_i(i(\omega t \pm i\eta + \Xi_{\alpha\beta}))$$

Exercise #3:  
(pg 401)

(7.26) "Quasi-particle Greens Function"

$$G_a^+(\omega) = \frac{1}{\omega^+ - \xi_a - \Sigma(\omega^+)} \\ = \frac{1}{\omega^+ - \xi_a - \Sigma'(\omega^+ - i0^+)}$$

(pg 401) "Green Function"

$$G^+(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} \cdot G^+(\omega) \\ = \frac{1}{2\pi} \int d\omega \frac{e^{-i\omega t}}{\omega - \xi_a} \\ = \frac{e^{-i(\xi_a + \Sigma')t + t\Sigma''}}{2\pi} \cdot F_i(i t (\omega - \xi_a - \Sigma'(\omega^+))) \\ \approx e^{-i t (\xi_a + \Sigma') + t \Sigma''} \cdot \Theta(t)$$

$$\text{when } \Theta(t) = \frac{F_i(i t (\omega - \xi_a - \Sigma'(\omega^+)))}{2\pi}$$

Exercise #4:  
(pg 403)

(7.30) "Spectral Density Function"

$$A_\alpha(\omega) = -2 \frac{\Sigma''(\omega)}{(\omega - \xi_\alpha - \Sigma'(\omega))^2 + (\Sigma''(\omega))^2}$$

(7.29) "Cauchy function"

$$A(\omega) = -2 \operatorname{Im} C^\dagger(\omega)$$

Normalization:

(from exercise) "Normal"

$$\int \frac{d\omega}{2\pi} A_\alpha(\omega) = \int \frac{d\omega}{2\pi} \left( -2 \frac{\Sigma''(\omega)}{(\omega - \xi_\alpha - \Sigma'(\omega))^2 + (\Sigma''(\omega))^2} \right)$$

$$= \frac{-i}{\pi} \int_{-\infty}^{\infty} d\omega C^\dagger(\omega)$$

$$= -\frac{i}{\pi} \circ 2\pi i \circ \frac{1}{2}$$

Cauchy's  
Residue  
Theorem

$$= 1$$

Exactness of Spectral Density:

(7.29) "Lehman Representation"

$$A(\omega) = \frac{2\pi}{Z} \sum_{\alpha\beta} \chi_{1\alpha\beta} \chi_{2\beta\alpha} \left[ e^{-\beta \Xi_\alpha} \xi e^{-\beta \Xi_\beta} \right] \delta(\omega + \Xi_{\alpha\beta})$$

$$\begin{aligned}
 \int \frac{d\omega}{2\pi} A_{\alpha}(\omega) &= \int \frac{d\omega}{2\pi} \left( \frac{2\pi}{z} \sum_{\alpha\beta} X_{1\alpha\beta} X_{2\beta\alpha} \left[ e^{-\beta E_{\alpha}} e^{-\beta E_{\beta}} \right] \right. \\
 &\quad \left. \cdot \delta(\omega + E_{\alpha\beta}) \right) \\
 &= \delta(\omega - \omega - E_{\alpha\beta}) \\
 &= 1
 \end{aligned}$$

Positive terms:

(pg 400)  $\Sigma(\omega^+)$  is analytic continuation of the function  $\Sigma(z)$  to the real axis"

(7.27) "Self-energy"

$$\Sigma'(\omega^+) = +\Sigma'(\omega), \quad \Sigma''(\omega^+) = -\Sigma''(\omega) < 0$$

So,  $\Sigma''(\omega) > 0$  and  $\Sigma'(\omega)$  is positive.

Exercise #5:

(pg 410)

(7.41) "Current"

$$j^\mu(x) = \int_{t' < t} dx' \cdot K^{\mu\nu}(x, x') A_\nu(x')$$

Gauge Transform

$$K^{\mu\nu}(x, x') \rightarrow K^{\mu\nu}(x, x') + \partial K$$

$$A_\nu(x') \rightarrow A_\nu(x') + \partial A_\nu$$

$$= \int_{t' < t} dx K^{\mu\nu}(x, x') A_\nu(x')$$

$$+ \int_{t' < t} dx' \cdot \partial_x K^{\mu\nu}(x, x') \cdot \partial_\nu A_\nu(x')$$

+ 0

+ ...

(pg 409) "Longitudinal conductivity"

$$j_i(\omega) = \sigma(\omega) \cdot E_i(\omega)$$

$$\sigma(\omega) = \frac{j_i(\omega)}{E_i(\omega)}$$

$$= \frac{\int dx K^{\mu\nu}(x, x') A_\nu(x') + \int dx \partial K^{\mu\nu} \partial A_\nu}{E_i(\omega)}$$

$\neq 0$ , not gauge

invariant.

(5.49) "Second London Equation"

$$\begin{aligned} j(r) &= \frac{\sigma}{\delta A(r)} \int d^d r \frac{n_s}{2m} A^2 \\ &= \frac{n_s}{m} A(r) \end{aligned}$$

If  $\nabla \cdot A = 0$ , then  $\nabla \cdot j(r) = 0$

Also,  $\nabla \times H = 4\pi j$  implies Meissner



effect when near zero.

(Pg 204) A bulk conductor  
cannot accommodate a  
magnetic field"



## Exercise #6:

(pg 411)

$$j = \sigma X$$

Current = Conductance  $\times$   
positive sign

Cite: Drude, P.

"Zur Elektronentheorie der  
Metalle; II. Teil. Galvano-  
magnetische und thermo-  
magnetische Effect"

University of Leipzig (1900)

(from problem) "Equation of Motion"

$$m \ddot{r} = -E - \frac{m}{\tau} \dot{r}$$

Force = Electrical Force - Frictional Force

Homogenous Second Order Linear  
Differential Equation:

① Equation:

$$\ddot{r}(t) + \frac{\dot{r}(t)}{\tau} + \frac{E}{m} = 0$$

② Roots to Equation of constant  
coefficients:

$$\ddot{r}(t) + \frac{\dot{r}(t)}{\tau} = -\frac{E}{m}$$

$$\lambda^2 + \frac{\lambda}{\tau} = \lambda(\lambda + \frac{1}{\tau})$$

$$= 0, \lambda_1 = 0, \lambda_2 = -\frac{1}{\tau}$$

③ Solution to Homogenous equation:

Summands:

$$r = \sum_{n=1}^{\infty} A_n(t) \cdot e^{\alpha t} \cdot \sin(\beta t) + B_n(t) \cdot e^{\alpha t} \cdot \cos(\beta t)$$

where  $\lambda = \alpha + i\beta$

$$\lambda_1 : \alpha = 0, \beta = 0$$

$$\lambda_2 : \alpha = -\frac{1}{\tau}, \beta = 0$$

$$r(t) = B e^{-\frac{t}{\tau}} + A$$

④ Solution to Particular equation:

Method of undetermined coefficients:

$$r = t^s \cdot e^{\alpha t} (C \cos(\beta t) + D \sin(\beta t))$$

$$\lambda_1 : \alpha = 0, \beta = 0, s = 1$$

$$r(t) = C \cdot t$$

$$r'(t) = C$$

$$r''(t) = 0$$

$$\ddot{r} + \frac{1}{\tau} \dot{r} = -\frac{E}{m}$$

$$\frac{1}{\tau} \cdot C = -\frac{E}{m}$$

$$C = -\frac{E}{m} \tau$$

$$r(t) = -\frac{E}{m} \tau$$

⑤ General Solution = Particular + Homogenous

$$r(t) = B \cdot e^{-\frac{t}{\tau}} - \frac{E}{m} \tau \cdot t + A$$

⑥ Boundary Conditions:

$$r(t=0) = 0 : r(0) = B \cdot e^{-0} - \frac{E}{m} \tau \cdot 0 + A = 0, B = A = 0$$

$$r(t) = -\frac{E}{m} \tau \cdot t$$

Velocity:

$$\begin{aligned} v(w) &= \frac{dr(t)}{dt} \\ &= -\frac{E}{m} \tau \end{aligned}$$

$$= -\frac{E}{m\omega}$$

Current:

(from problem) "conductance"

$$\sigma(\omega) = \frac{n}{m} \cdot \frac{1}{\tau - i\omega}$$

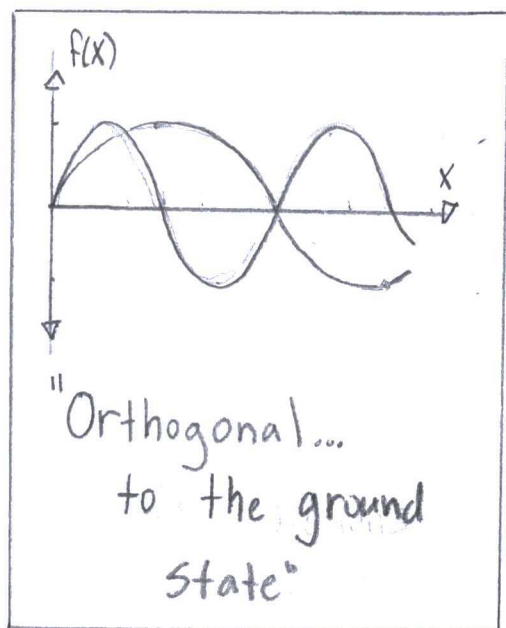
(from problem) "Drude Current"

$$\vec{j} = -n \cdot \vec{v}$$

$$= -n \cdot \left( -\frac{E}{m} \tau \right)$$

$$\text{Re}(j) = \sigma E \quad , \text{ when } \omega \ll 1$$

## Problem 7.6.1 - Orthogonality Catastrophe:



a) (from problem) "Spectral Density"

$$A_{nm} = \langle n | m' \rangle$$

(from problem) "Susceptibility"

$$\chi \equiv \det(A)$$

(from problem) "States"

$$\sum_{\epsilon_{m'} > E_F} |\langle n | m' \rangle|^2 \ll 1$$

$$\text{So, } \ln(\chi) = \ln(\det(A))$$

$$= \ln\left(\sum_{\epsilon_{m'} < E_F} |\langle n | m' \rangle|^2\right)$$

$$= \ln\left(1 - \sum_{\epsilon_{m'} > E_F} |\langle n | m' \rangle|^2\right)$$

$$= - \sum_{\epsilon_{m'} > E_F} |\langle n | m' \rangle|^2$$

$$= -L$$

Logarithm  
Expansion:

$$\ln(1-x) = -x + \dots$$

$$\chi = e^{-L}$$

b) If  $A_{nm} = \langle n|m \rangle$

$$= \int_0^{2\pi} \int_0^\pi \int_0^R d\phi d\theta dr \cdot \phi_n(r) \cdot \phi_m(r) \cdot r^2 \sin\theta$$

(from problem) "Unperturbed state"

$$\phi_n(r) = \frac{N_n \sin(k_n r)}{k_n \cdot r}$$

(from problem) "Perturbed state"

$$\phi'_n(r) = N_n \cdot \sin(k_n \cdot r + \delta_m (1 - \frac{r}{R})) \cdot \frac{1}{k_n \cdot r}$$

$$= 4\pi N_n N_m \int_0^R dr \cdot r^2 \frac{\sin(k_n r)}{k_n \cdot r} \frac{\sin(k_m r + \delta(1 - \frac{r}{R}))}{k_m \cdot r}$$

$$= \frac{2\pi N_n N_m}{k_n k_m} \frac{\sin \delta_m}{k_n - k_m + \frac{\delta_m}{R}}$$

(From problem)  
"Normalization"

$$N_n = \frac{k_n}{\sqrt{2\pi R}}$$

$$= \frac{2\pi}{k_n k_m} \frac{k_n}{\sqrt{2\pi R}} \frac{k_m}{\sqrt{2\pi R}} \frac{\sin \delta_m}{k_n - k_m + \frac{\delta_m}{R}}$$

$$= \frac{\sin \delta_m}{\pi(n-m) + \delta_m}$$

... Where  $k_n = \frac{\pi n}{R}$



## Problem 7.6.2 - RPA dielectric function

a) (7.31) "Spectral function weighted by Fermi or Bose distribution function"

$$\int \frac{d\omega}{2\pi} n_{F/B}(\omega) A_n(\omega) = \langle \hat{n}_\alpha \rangle$$

Equation 7.34 Derivation:

(4.33) "Dielectric Function"

$$\epsilon(q) = 1 - V(q) \Pi_q$$

(4.29) "Polarization Operator"

$$\Pi_q = \frac{2T}{L^3} \sum_p G_p \cdot G_{p+q}$$

(7.34) "Dielectric function"

$$\begin{aligned} \epsilon(q, \omega)^{-1} &= 1 - \frac{V_0(q)}{L^d} \int d\tau e^{i\omega_n \tau} \langle \hat{n}(q, \tau) \hat{n}(-q, 0) \rangle \\ &= 1 - \frac{V_0(q)}{L^d} \int d\tau \cdot e^{i\omega_n \tau} \frac{A_n(\omega)}{2\pi(e^{\beta\hbar\omega} \pm 1)} \end{aligned}$$

(7.32) "Cauchy Theorem"

$$C(z) = \int_{-\infty}^{\infty} \frac{dw}{2\pi} \frac{A(w)}{z - w}$$

$$\epsilon(q, w)^{-1} = 1 - \frac{V_0(q)}{L^d} \int d\tau \cdot e^{i w \tau} \cdot C(z)$$

$$= 1 - \frac{V_0(q)}{L^d} C^+(q, w)$$

where  $z = e^{\beta \Delta E}$

b) As an equation:

$$\epsilon_q^{-1} = 1 - \frac{T V_0(q)}{L^d} \langle n_q n_{-q} \rangle_c$$

$$= 1 - V_0(q) \frac{\chi_0(q)}{1 + V_0(q) \chi_0(q)}$$

$$= \frac{1}{1 + V_0 \chi_0(q)}$$

$$= \frac{1}{1 - \frac{T V_0(q)}{L^d} \langle n_q n_{-q} \rangle_{irr}}$$

$$\text{So, } \epsilon_q = \left( 1 - \frac{T V_0(q)}{L^d} \langle n_q n_{-q} \rangle_{irr} \right)^{-1}$$

$$= 1 + \frac{T V_0(q)}{L^d} \langle \hat{n}_q \hat{n}_{-q} \rangle_{irr}$$

As a diagram:

$$\epsilon_q^{-1} = 1 - \frac{T V_0(q)}{L^d} \langle n_q n_{-q} \rangle_c$$

$$= 1 - \frac{\text{diagram 1}}{1 + \text{diagram 2}}$$

Diagram 1: A wavy line connected to a loop with two arrows pointing in opposite directions.

Diagram 2: A wavy line connected to a loop with two arrows pointing in the same direction.

$$= 1 - \text{diagram 3}$$

Diagram 3: A wavy line connected to a shaded loop with two arrows pointing in opposite directions.

$$= 1 - \frac{V_0 \chi_0(q)}{1 + V_0(q) \chi_0(q)}$$

$$= \frac{1}{1 + \frac{T V_0(q)}{L^d} \langle \hat{n}_q \hat{n}_{-q} \rangle}$$

$$= \frac{1}{1 + \text{diagram 2}}$$

Diagram 2: A wavy line connected to a loop with two arrows pointing in the same direction.

Problem 7.6.3 - Electromagnetic Response of Quantum Dots:

a) (5.62) "Dissipative Tunneling Action"

$$S[\phi] = \int d\tau \left( \frac{\dot{\phi}^2}{4E_C} - iN_0 \dot{\phi} \right) - g_T \int d\tau d\tau' \frac{\sin^2(\frac{1}{2}(\phi(\tau) - \phi(\tau'))}{\sin^2(\pi T(\tau - \tau'))}$$

$$S_u[\phi] = S[\phi] - iV(\tau)$$

$$= \int d\tau \left( \frac{V^2}{4E_C} - iN_0 V \right) - \text{tr} \ln \begin{pmatrix} 2\tau - \mu + i(V + \epsilon_d) & T \\ T^\dagger & 2\tau - \mu + \epsilon_1 \end{pmatrix}$$

$$S^{(2)}[\phi, V] = \frac{1}{4E_C T} \sum_{\tau} \omega_m^2 |\phi_m|^2 + \frac{g_m}{4\pi T} \sum_m \sin^2\left(\frac{1}{2}\Delta\phi\right)$$

$$= \frac{1}{4E_C T} \sum_{\tau} \omega_m^2 |\phi_m|^2 + \frac{g_m}{4\pi T} \sum_{\tau} |\omega| \left| \phi + \frac{V}{im} \right|^2$$

Linear Response Kernel:

(7.43) "Response"

$$K^{\mu\nu}(x, x') = \frac{1}{Z} \frac{\delta^2}{\delta A_\mu(x) \delta A_\nu(x')} Z[A]$$

$$= \frac{1}{Z} \frac{\delta^2}{\delta U \delta U} Z[U]$$

$$= -\frac{g_T |\omega|}{2\pi T \omega^4} (-\omega^2 + \frac{g_T |\omega|}{2\pi T} (\langle \phi \omega \rangle - iU)^2)$$

$$\text{Re}(K^{\mu\nu}(x, x')) = \frac{g_T}{2\pi T \omega^2} \left( 1 - \frac{g_T |\omega|}{2\pi T} \langle \phi \omega \rangle \right)$$

$$= \frac{g_T}{2\pi T \omega^2} \left( 1 - \frac{g_T |\omega|}{2\pi T} \frac{1}{\frac{\omega^2}{2E_C T} + \frac{g_T |\omega|}{2\pi T}} \right)$$

$$= - \frac{1}{\frac{2\pi |\omega|}{g_T} + 2E_C}$$

### Currents

When  $I = i 2\pi \delta N$

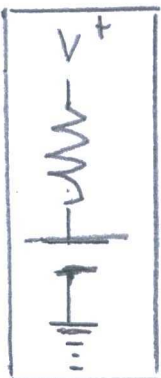
$$= i (K_{00})_m U_m$$

$$= \frac{U_m}{\frac{2\pi |\omega|}{g_T} + 2E_C}$$

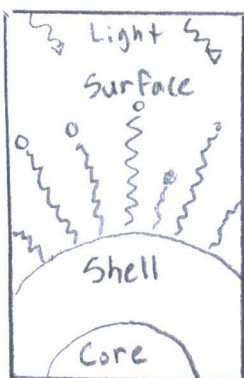
$$= \frac{U_m}{\frac{2\pi}{g_T} - \frac{2E_C}{i\omega}} \quad \dots \text{as } |\omega| \rightarrow i\omega$$

### b) Kirchoffs Law

Circuit



Quantum Dot



Surface :  $V = I_\omega R$  (Ohm's Law)

$$= U + IR \quad (\text{Bias})$$

Shell :  $= U + \frac{I}{i\omega C} \quad (\text{Impedance})$

$$I = \frac{V}{R - \frac{1}{i\omega C}}$$

$$\propto \frac{V}{\frac{2\pi}{9T} - \frac{2E_c}{i\omega}}$$

Circuit Equation  
relates to  
Response Function

Top concepts relating to response in materials:

1. Fourier Transform

2. Kramer-Kronig relation.

3. Spectral Density.